randomized controlled trials. They concluded that deactivating Facebook increases subjective well-being by about 25-40% as much as standard psychological treatments, a key finding Meta cites in a presentation related to the potential harm it is causing in 2019.[393]

383. For reasons that are not entirely clear, Mathew Gentzkow of Stanford ("an exceptionally acclaimed Stanford economist," per Brent Harris[394]), an author on the Alcott et al "deprivation" experiment, was in contact with people at Meta prior to the publication of the first study alerting them to the findings.[395] Within Meta, there was concern about openly collaborating with him and his coauthors since there were, according to an email from John Hegeman, "risks about what [the study] might find and … the substantial corresponding PR of those findings."[396] David Ginsberg, who started as a Director of Research and eventually became Vice President of Global Communications and Public affairs,[397] replies that he is favor of the study in principle, but given the risk, he is "not convinced [they] need or should do it with external partners."[398]

384. Rather than participate in this one, the suggestion is made to launch their own study fronted by Nielsen so as to mask their involvement in it. But Annie Franco, another Meta researcher, responds that while they should do their own study, if the Stanford groups is negative and Meta's is positive they will have a hard time convincing people that their results are credible.[399] In spite of that concern, Meta decides to launch its own internal study called the "Nielsen thing." In the meantime, the Alcott study was completed and shared with Meta researchers.

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[393] Haugen_00010114, -0123-24
[394] *See* META3047MDL-014-00048189, -8189
[395] META3047MDL-040-00586960 at -6965
[396] META3047MDL-040-00590304 at -0308
[397] Ginsberg Dep. at 53-55
[398] META3047MDL-040-00590304 at -0307
[399] META3047MDL-040-00590304 at -3087

385. Alex Leavitt, a lead quantitative UX researcher at Meta, was impressed with the Alcott study, responding to the abstract "we think the paper is great, and we're excited to learn more from it and help our teams understand the implications.[400] But they are worried that the press "will inflate the narrative." Later, Meta tried to get them to recast their relative effect size to make it not seem as substantive.[401]

386. Ultimately, Gentskow gave leaders at Meta a heads up that their paper would be in the *New York Times*.[402] The press that Alcott's study received further motivated Meta to advance the Nielsen project.[403] They ultimately decided to pilot the Nielsen study internally and then partner with academics "more fully afterwards."[404] However, even before it launched a Meta memo contained the following caveat:

> Publication / External communication: There are concerns about publishing our final report, and even talking about this study externally. There were even fears of running this study and what will we do if we find out negative things about Facebook (comparisons to cigarette studies). We had to get this study reviewed all the way up to Nick and Javi.

Document 122: META3047MDL-019-00095647 at -9569

387. In fact, the sensitivity to potential leaks of the findings or the study itself were great enough that being employed in journalism was an exclusion criteria for the study.[405] As the study ensues, there is this notable exchange amongst well-being scientists in which they discuss whether or not the findings will be made public:

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400 META3047MDL-040-00586960 at -6966
401 META3047MDL-040-00586960 at -6964
402 META3047MDL-014-00048189
403 META3047MDL-003-00176564 at -6568
404 META3047MDL-019-00095647 at -5648
405 META3047MDL-020-00572432 at -2435

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[redacted](9/19/2019 15:43:54 PDT):
>so if we get the nielsen results back, if they are good would we publish or is there no intention of publishing?

[redacted](9/19/2019 15:45:09 PDT):
[redacted] - we just aren't planning on it

[redacted](9/19/2019 15:49:36 PDT):
>ok. and just curious, do we think the results will be that negative? ive got to think that even if slightly negative they wont be THAT bad...i feel like all the well being research ive seen if it says FB isnt good for people it's always jsut like a smallllllll percentage over neutral

[redacted](9/19/2019 15:49:52 PDT):
[redacted]

[redacted](9/19/2019 15:49:55 PDT):
[redacted]im being naive but i feel like that this will bbe fine

[redacted](9/19/2019 15:50:01 PDT):
>I think we'll see some positive

[redacted](9/19/2019 15:50:06 PDT):
[redacted]

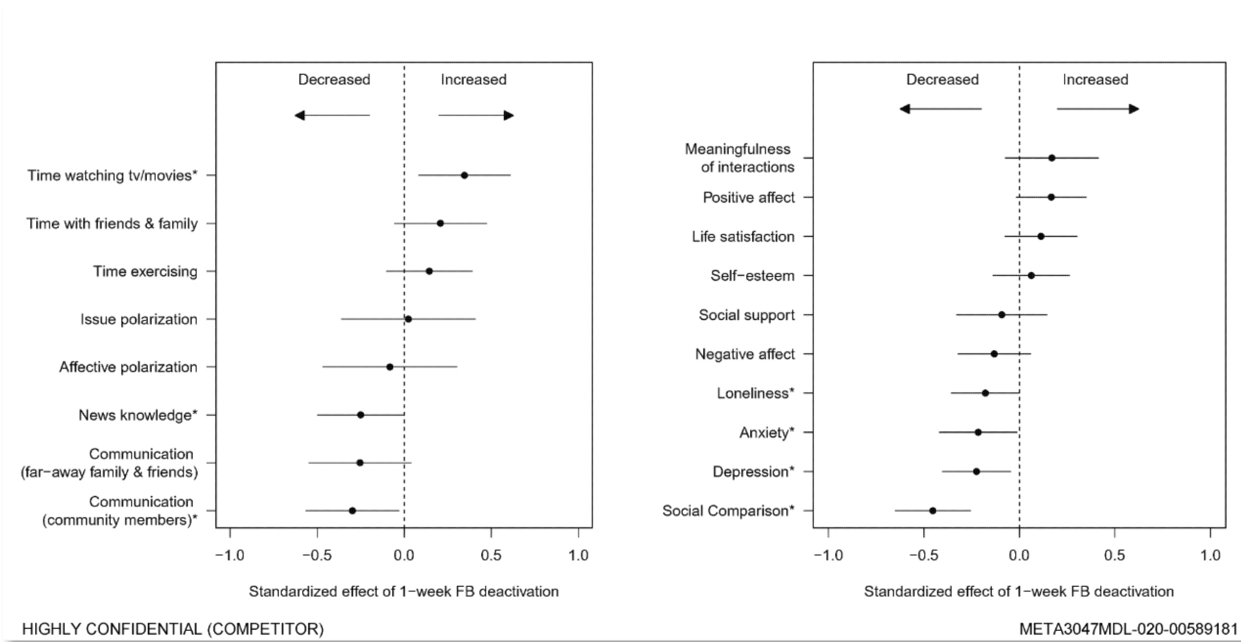
[redacted](9/19/2019 15:50:06 PDT):
[redacted]

[redacted](9/19/2019 15:50:10 PDT):
>and most neutral
```

Document 123: META3047MDL-040-00031943 at -1943

388. Later in this same exchange, [redacted] (News Feed and Algorithmic Transparency Communications Manager)[406] responds "someone just said to me if the results are bad and we don't publish and they leak is it going to look like tobacco companies doing research and knowing cigs were bad and then keeping that info to themselves." [redacted] (User Experience Researcher) attempts to reassure him "no, because we will do something about it." The results of the "pilot" are presented below.

[406] https://www.linkedin.com/in/james-raimo-64180b102/



Document 124: META3047MDL-020-00589181 at -9181

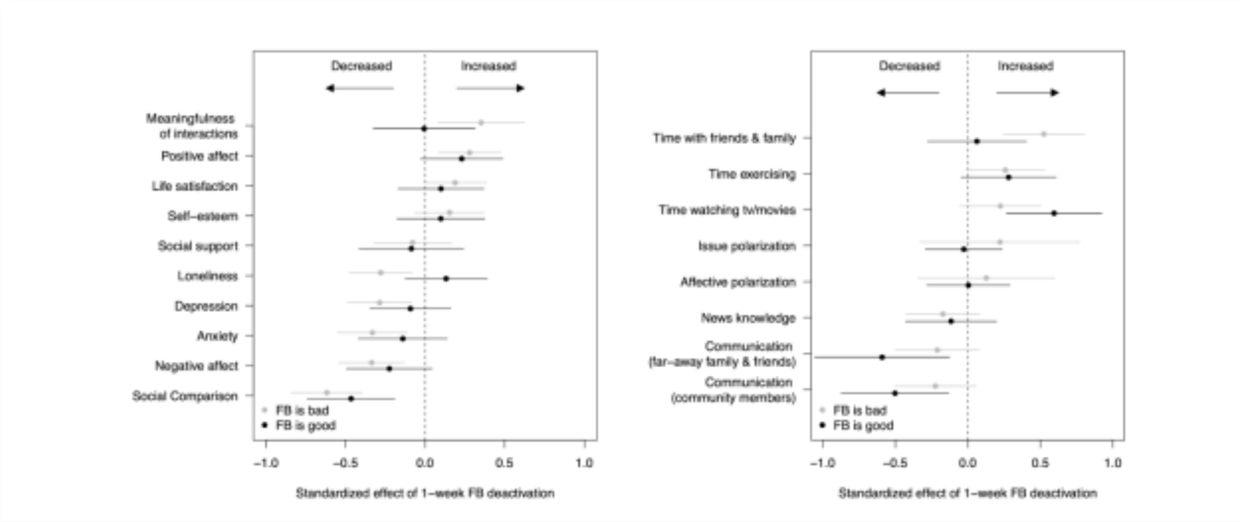
389. After 1 week (which in prior studies was too soon to see effects and sometimes even found withdrawal effects), most of the findings are in the direction that is unfavorable for Meta, and some of the results including less social comparison, depression, anxiety and loneliness, were statistically significant. Their own internal summary was as follows:

> We found that those who stopped using Facebook for a week:
>
> - Reported lower feelings of depression, anxiety, loneliness, and social comparison
> - Reported fewer interactions with community members
> - Reported more time watching tv/movies and spent less time on social media
> - Had lower news knowledge

Document 125: META3047MDL-047-00058006 at Slide 2

390. These results are consistent with Alcott's study as well as with numerous other longitudinal and experimental studies and the differences were evident after only one week. But the team decided subsequently to then stratify the results to see what role "expectations" played in

the observed results. To do this, they asked people at enrollment if they think Facebook is bad for their well-being and compared the results for those that believed it was and those that believed it did was not. Here is what they found in that stratified analysis:



Document 126: META3047MDL-047-00058006 at Slide 7

391. Conditioning on whether users believed Facebook was bad for them considerably attenuated the effect sizes except for social comparison which remained the same. Notably, many of the effects are still in the same direction but are no longer statistically significant, and the fact that social comparison was unaffected was itself an important finding. Their read of the results, that expectations drove outcomes, presupposes that the findings were based entirely on people being primed to believe that Facebook worsened well-being. But of course, there is an alternative explanation which is that some people rightly deduced that for *them*, Facebook was harmful. That is, they had an accurate assessment of its effect on them. This is completely consistent with the differential susceptibility theory and "addiction theory." Quitting drinking for example benefits those who think it's a problem for them more than those who do not because people who think they have a drinking problem generally do.

392. Later, Paul Dow writes to Moira inquiring what happened with the study:

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Paul Alexander Dow (5/04/2021 12:09:11 PDT):
>hey. you might have told me this, but, if so, i forgot. what ever happened with the Nielsen deprivation
study? I think I lost track when covid hit. I know there was one or more pilot studies completed, but did
they ever do the full thing?

Moira Kathleen Ballantyne Burke (5/04/2021 12:12:07 PDT):
>There were two one-wave pilot studies and the findings have been shared with leads internally, but afaik
we're not planning to do a longer-term study. The findings were basically that people's expectations
totally drove the results, and we weren't able to use the data to disentangle those expectation effects.
Summary:  https://fb.quip.com/MkxAA3gfY84C

Paul Alexander Dow (5/04/2021 12:29:55 PDT):
>well, shit. that seems like a result worth publicizing on its own. not that anyone would want to hear it
from us. maybe that is a study for FORT, what is the role of expectation effects on social media
deprivation studies. i see Safaa has a deprivation study in her list of research projects for the
Problematic Use stream.
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Document 127: META3047MDL-040-00213905 at -3905

393. The fact that they could not disentangle the effects especially given that social comparison remained significant is hardly justification for not publishing (or trying to publish) the results. In fact, if they believed their results were robust, they should have subjected their assertion that the observed effects were driven by the lack of blinding, to peer review and then made them public. As [redacted] says in an email on October 23, 2020:

The **main findings** of the study are:

1. People who stopped using Facebook for a week reported an improvement in certain types of emotions (e.g., depression, anxiety, loneliness, and social comparison), but no change in other outcomes (e.g., political polarization, or meaningful interactions).
2. Most of the effects we found are linked to people's beliefs about Facebook. For example, the reductions in feelings of depression, anxiety, and loneliness associated with deactivation appeared mostly among participants who believed (prior to participating in the study) that Facebook was bad for them.

The main implication of our findings: External criticisms highlighting the negative impact of the company products on well-being related outcomes may be overblown. For example, if beliefs are the result of the existing media narrative around the company, the findings in the Nielsen study, and similar external work, will overestimate the true effect of Facebook (akin to placebo effects).

Document 128: META3047MDL-165-00000428 at -0429

394. Their interpretation of their findings is, scientifically speaking, what is "overblown". A post hoc analysis showing that the effects were larger for those who believed that they had a problem at enrollment hardly invalidates the findings. If anything, it is consistent with

other studies that found that people with baseline mental health symptoms or problematic usage benefit more from diminishing or eliminating usage. He goes on to say that he is "working with Comms to determine the timing and framework under which we would make the study public if we chose to do so." Given that they have not (yet) released it, one can deduce that they have buried it and perhaps have mimicked the tobacco companies as some of them feared they might.

395. Recently, Allcott, Gentzkow et. al. repeated this seminal SM deprivation study with some notable differences. First, among the 27 authors, there are 7 affiliated with Meta. Second, related to that collaboration, deprivation in this study was accomplished by central deactivation of the account by Meta (rather than voluntary abstinence). Third, the study included both Instagram and Facebook (more on that later) whereas the prior one was limited to Facebook alone. Fourth, they collected passive sensing data on a subset of participants which enabled them to track if people spent additional time offline or on other apps. Finally, they randomized participants to deactivate for 1 week vs. 6 weeks. This differed from the prior study in two important ways: it was longer in duration and better blinded since both groups had reason to believe that they were in the intervention arm.

396. The scale of the study was impressive and the design was incredibly robust. They recruited over 19,000 Facebook users and 15,000 Insta users making it by far the largest publicly available experiment of SMS use and they attempted to address demand effects. However, before delving into the findings, two important attributes of the study warrant mentioning. First, only 1% of people offered the opportunity to participate in the study ultimately did. This has implications for the generalizability of the findings which the authors acknowledge. It's worth considering which way(s) participants might differ from non-participants. The most obvious one is that they are willing to "deactivate" their account which likely means they are less likely to be compulsive

or problematic users. By analogy, a study recruiting people to participate in dry January (or not) would assuredly under enroll problematic drinkers. Second, participants were free to use other apps as they wished and the findings suggest that in the Instagram group, they effectively replaced time on that app entirely with time on others (including Snap, YouTube, and TikTok) which would attenuate the findings insofar as those social media sites have many similar features and effects.

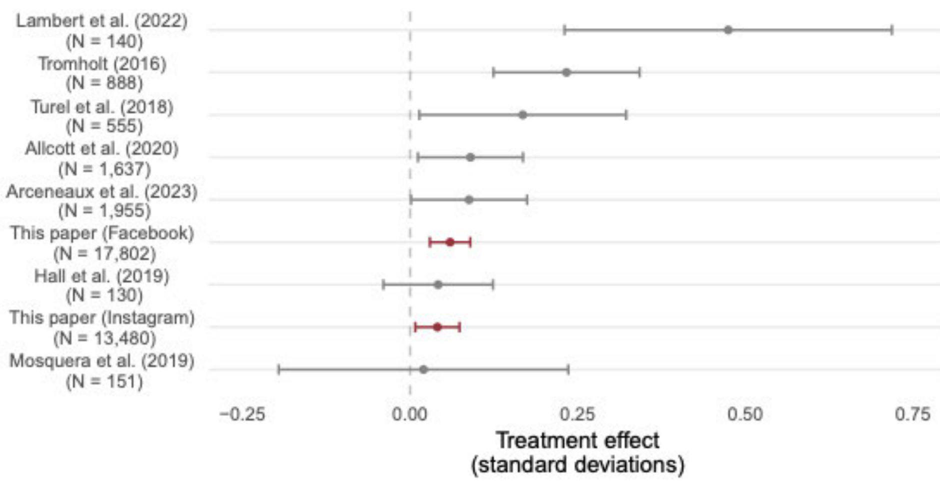
397. For the purposes of this report, I will summarize the Instagram results since they are most relevant to the plaintiffs in the case. Deactivation of Instagram resulted in a .041 standard deviation improvement in the emotional state index. In a sub-analysis, the effect was largest for women 18-24 with a .11 standard deviation improvement. The authors contextualize these effects by comparing them to those found in a metanalysis of 419 randomized trials of nine types of psychological interventions which improve subjective well-being by .16 to .42 standard deviations.[407] Seen in this light, deactivation of Instagram is between 10 and 25% as effective as psychotherapy for all people and between 25 and 69% as effective for young women. However, from a public health standpoint, given that ready access to psychotherapy is both infeasible and unscalable, the overall effects of SM reduction on population health are likely considerably greater.

398. The authors of the paper go further and contextualize their findings compared to other experimental studies of SM deprivation with durations great than one week. Their summary represents an independent collection of relevant experiments as acknowledged by the Meta employees included as authors. Those results are presented in their Figure 3 below:

[407] Van Agteren J, Iasiello M, Lo L, et al. A systematic review and meta-analysis of psychological interventions to improve mental wellbeing. Nature Human Behaviour. 2021/05/01 2021;5(5):631-652. doi:10.1038/s41562-021-01093-w

Figure 42: Summary of Relevant Experiments Externally Published and Known to Meta

Figure 3: Comparison to Other Experimental Estimates



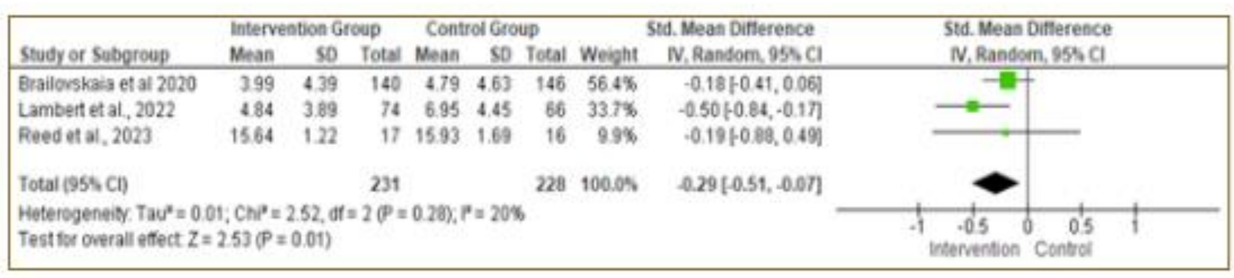
Note: This figure compares our Facebook and Instagram estimates with other experimental results of social media deactivation by Tromholt (2016), Turel, Cavagnaro and Meshi (2018), Allcott et al. (2020), Mosquera et al. (2020), Hall et al. (2021), Lambert et al. (2022), and Arceneaux et al. (2023). For each paper, we compute treatment effects on the paper's subjective well-being outcomes, in units of standard deviations of the outcome variable in the respective study's sample.

Notably, every study had positive effects and 7 of 9 were statistically significant. The magnitude of the effect in this study was smaller than most others.

399. One final point regarding this study warrants mentioning. It is error to claim that the only data that can establish causality must come from experimental studies. This, the largest and best experimental study done to date, highlights two salient problems with that contention. First, given low participation rates, it will be difficult to generalize from experiment participants to the overall population. Second, given substitution effects, deprivation studies of one site will measure the net effect of reducing use of one SM site while increasing the usage of others. In my opinion, both of these limitations make the observed effects conservative and argue for the consideration of all reliable studies in trying to determine causality.

400. A metanalysis assessed the effects of experimental evaluations of "digital detox" on mental health outcomes. Again, these studies tested the putative benefits of a period of abstinence from social media sites.[408] There was considerable and statistically significant heterogeneity in both the approaches and the outcomes (p<.000001). While the overall effect of "detox" on well-being was null (which is not surprising given the heterogeneity) the effects on depression was moderate and significant with an effect size of -.29 (see below).

Figure 43: Summary of Metanalyses of Experimental Evaluations of 'Digital Detox on Mental Health Outcomes



This study also neglected to include the Allcott experiment for unclear reasons which would have further enhanced the results and made them more robust given its findings.[409]

401. Meta itself conducted a study of 6,000 Instagram users to test the effect of social comparisons on affect. They did so by randomly sequencing questions related to overall well-being and those related to negative social comparison. Below is a schematic recreated for image clarity from Haugen_00000797, -0864.

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[408] Ramadhan RN, Rampengan DD, Yumnanisha DA, et al. Impacts of digital social media detox for mental health: A systematic review and meta-analysis. *Narra J.* Aug 2024;4(2):e786. doi:10.52225/narra.v4i2.786

[409] Marciano L, Schulz PJ, Camerini A-L. Cyberbullying Perpetration and Victimization in Youth: A Meta-Analysis of Longitudinal Studies. *Journal of Computer-Mediated Communication*. 2020;25(2):163-181. doi:10.1093/jcmc/zmz031